Project Design Phase-II

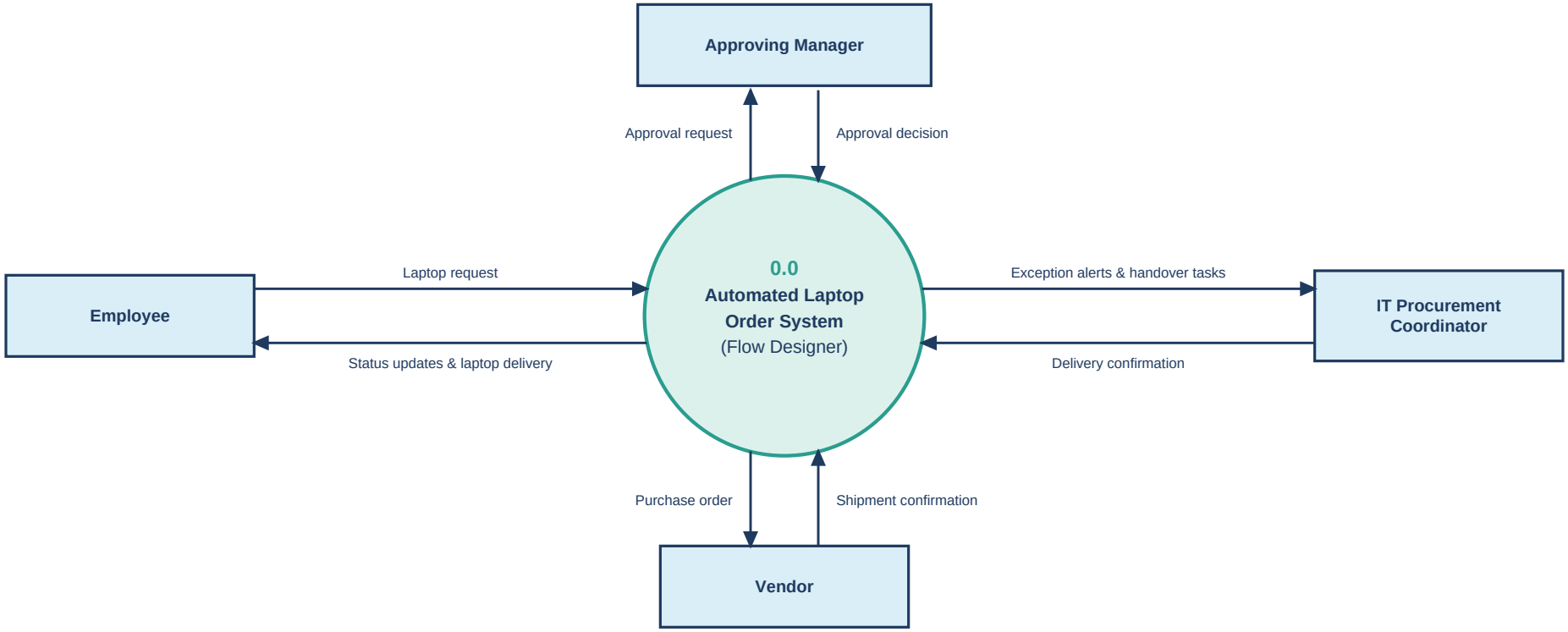
Data Flow Diagram & User Stories

Date	2 October 2026
Team ID	
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Project: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

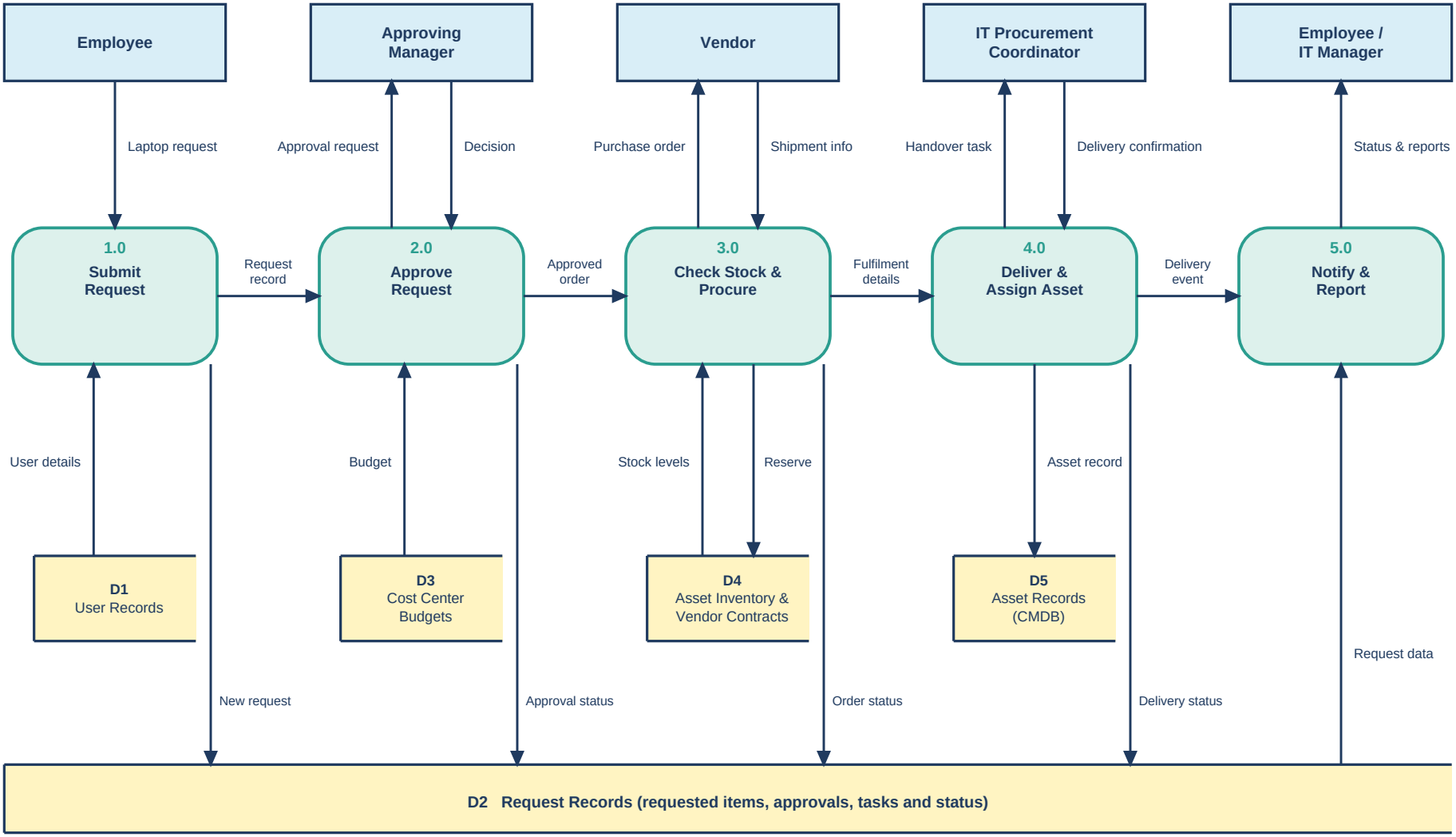
A Data Flow Diagram (DFD) is a visual representation of how information moves through a system. It shows where data comes from, which processes transform it, where it is stored and where it goes. It helps the team agree on the flow of data before development begins.

Data Flow Diagram - Level 0 (Context Diagram)



Reading the diagram: blue rectangles are external entities (people or organisations), the green shape is the process, and arrows show the data that moves between them. The whole laptop ordering flow is treated as a single process built in Flow Designer.

Data Flow Diagram - Level 1



Legend: blue rectangle = external entity | green rounded box = process | yellow open box = data store | arrow = data flow

Process Descriptions (Level 1)

Process	What the flow does	Data in	Data out	Data stores
1.0 Submit Request	Employee picks the standard laptop catalog item; details are auto-filled; eligibility and duplicate checks run.	Laptop request, user details	Validated request record	D1, D2
2.0 Approve Request	Skips approval for pre-approved roles; otherwise sends a one-click approval, checks the cost center budget and escalates after 2 days.	Request record, budget, decision	Approved order	D2, D3
3.0 Check Stock & Procure	Checks and reserves stock; raises a purchase order automatically when stock is low and assigns the vendor by region or price.	Approved order, stock levels	Fulfilment details, purchase order	D2, D4
4.0 Deliver & Assign Asset	Tracks shipment, confirms delivery, then creates the asset record and assigns it to the employee.	Fulfilment details, delivery confirmation	Asset record, delivery event	D2, D5
5.0 Notify & Report	Sends a notification at each stage, updates the portal status tracker and feeds the turnaround and savings dashboard.	Delivery event, request data	Status updates, dashboard	D2

Data Stores

ID	Data store	Contents
D1	User Records	Employee role, location, department and eligibility information
D2	Request Records	Requested items, approval status, tasks and stage history
D3	Cost Center Budgets	Budget available per cost center for laptop spend
D4	Asset Inventory & Vendor Contracts	Laptops in stock, reservations, vendor regions and contract prices
D5	Asset Records (CMDB)	Asset details and assignment to the employee

User Stories

Each story follows the pattern: *As a [user type], I want [goal], so that [benefit].* Priority and release follow the earlier idea prioritization: build first, build next, plan for later.

User Type	Functional Requirement (Epic)	USN	User Story / Task	Acceptance Criteria	Priority	Release
Employee	Request a laptop	USN-1	As an employee, I want to request a standard laptop from one catalog item with pre-filled specs, so that I do not have to enter details again.	Catalog item is visible on the portal; role, location and department are filled in automatically.	High	Sprint-1
Employee	Request a laptop	USN-2	As an employee, I want my eligibility checked and duplicate requests blocked, so that only valid requests are submitted.	Ineligible users see a clear message; a second request inside the refresh cycle is rejected.	High	Sprint-1
Manager	Approval	USN-3	As a manager, I want requests from pre-approved roles to skip my approval, so that I do not spend time on routine requests.	Pre-approved roles go straight to fulfilment; no approval task is created.	High	Sprint-1
IT Coordinator	Fulfilment	USN-4	As an IT coordinator, I want stock checked and a laptop reserved automatically, so that I do not check inventory by hand.	Inventory is checked on approval; a laptop is reserved, including for new hires before day one.	High	Sprint-1
Employee	Tracking	USN-5	As an employee, I want to be notified at each stage and see a status tracker, so that I know where my request is.	Notifications on approved, shipped and delivered; the portal shows the live stage.	High	Sprint-1
Manager	Approval	USN-6	As a manager, I want to approve by email or mobile in one click, with the budget shown, so that approvals are quick and within budget.	Approve or reject works from email or mobile; cost center budget check shows before approval.	Medium	Sprint-2
Manager	Approval	USN-7	As a manager, I want pending approvals escalated automatically, so that requests do not stall.	A request pending over 2 days is escalated and a reminder is sent.	Medium	Sprint-2
IT Coordinator	Procurement	USN-8	As an IT coordinator, I want a purchase order created automatically when stock is low, so that I do not raise orders manually.	A purchase order is created for the right model when stock falls below the threshold.	Medium	Sprint-2
IT Coordinator	Procurement	USN-9	As an IT coordinator, I want the vendor assigned by region or contract price, so that orders go to the best supplier.	The vendor is picked using region and contract price rules; the choice is stored on the order.	Low	Sprint-3
IT Coordinator	Asset management	USN-10	As an IT coordinator, I want the asset record created and assigned on delivery, so that records stay accurate.	On delivery confirmation, an asset record is created and assigned to the employee.	Low	Sprint-3
IT Manager	Reporting	USN-11	As an IT manager, I want a dashboard of turnaround time and cost savings, so that I can measure the benefit of automation.	Dashboard shows average turnaround, requests per stage and estimated savings.	Low	Sprint-3

Summary: 11 user stories across 3 releases. Sprint-1 (High) delivers the core request, approval-skip, stock check and notifications; Sprint-2 (Medium) adds one-click approval, escalation and automatic purchase orders; Sprint-3 (Low) covers vendor assignment, asset records and the dashboard.

Implementation Screenshots (ServiceNow)

Captured from the project demo video: Flow Designer build and a test laptop request.

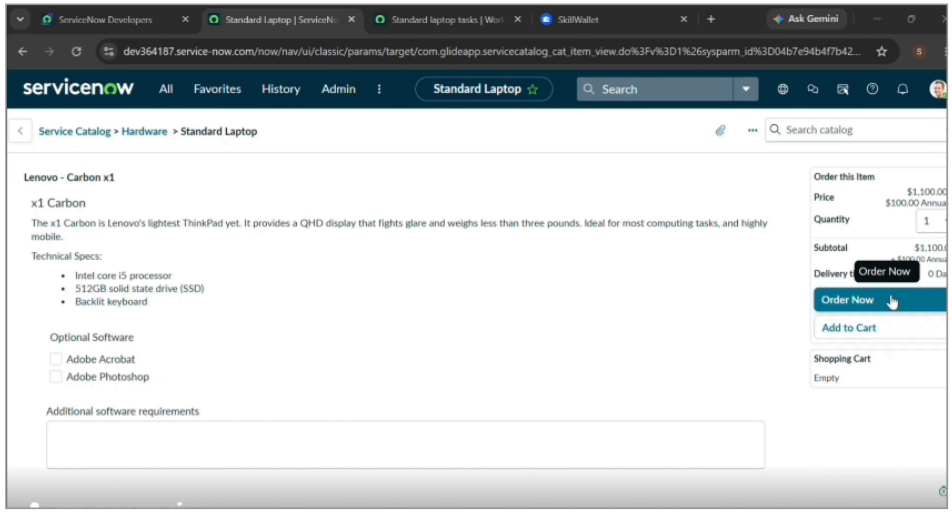


Figure 1: Standard Laptop order page with Order Now

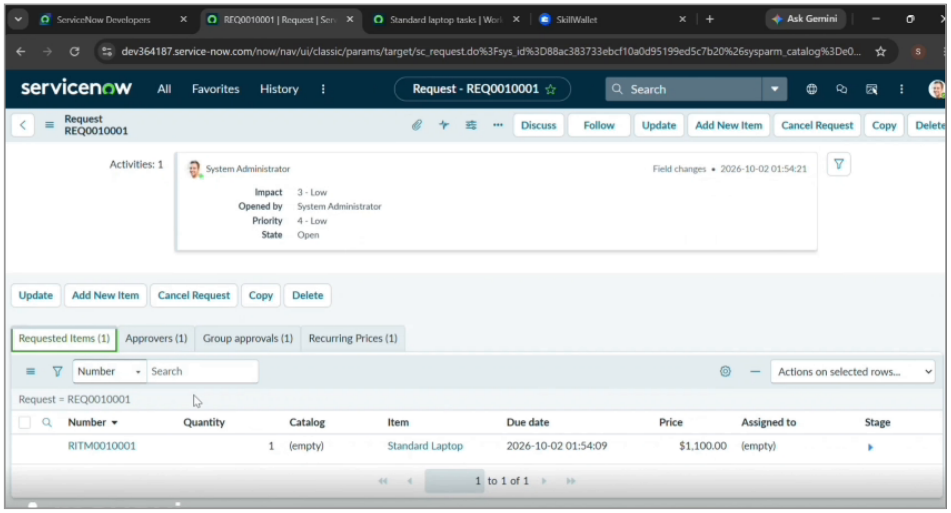


Figure 2: Request REQ0010001 with approvers and requested items

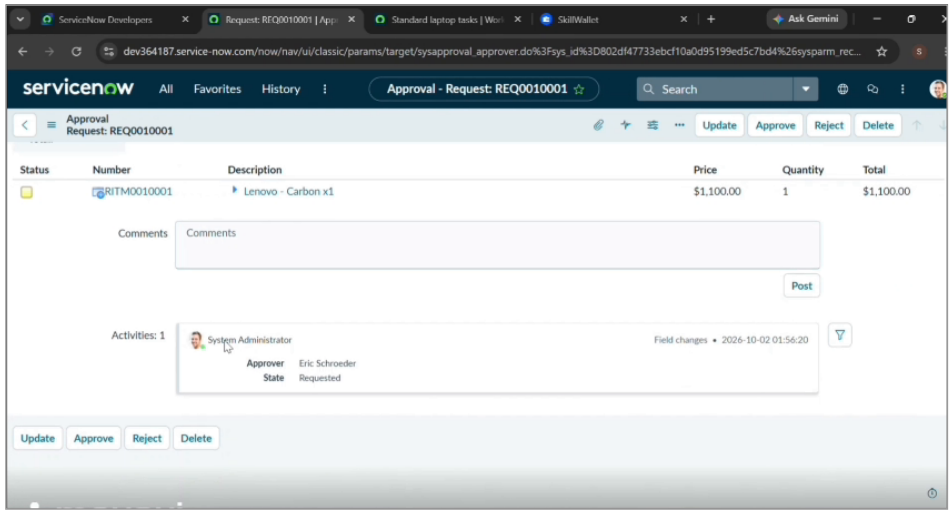


Figure 3: Approval record for REQ0010001 (Approve / Reject)

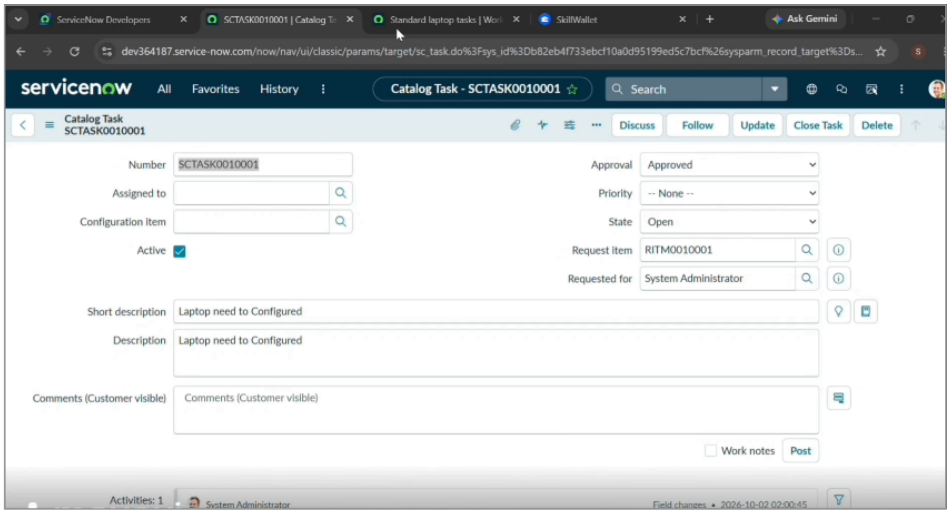


Figure 4: Catalog task SCTASK0010001 "Laptop need to Configure"